



Fig. 3: Mesh network and modifications

time taken by the scheduler to allocate bandwidth for the packet in queue.

IEEE 802.15.4 defines a total of three frequency bands of operation: 2.4GHz, 915MHz and 868MHz. The 2.4GHz band offers a total of 16 channels (channel 11-26), 915MHz offers 10 channels (channel 1-10), and 868MHz offers just one channel (channel 0). The bit rate of the protocol depends on the selection of frequency of operation.

Frequency	Bit rate	Channels - Range
2.4GHz	250kbps	16 (11-26)
915MHz	40kbps	10 (1-10)
868MHz	20kbps	1 (0)

The maximum length of an IEEE 802.15.4 MAC packet is 127 bytes. Each packet consists of header bytes and 16-bit cyclic redundancy check (CRC) value. The 16-bit CRC value verifies the frame integrity. In addition, it optionally uses an acknowledged data transfer mechanism. With this method, all frames with a special ACK flag set are acknowledged by its receiver.

IEEE 802.15.4 based WSNs can support a maximum of 250kbps for 2.4GHz bandwidth. Therefore WSNs are employed in areas requiring intrusion detection or fire alarm, and where the packets are generated at very low data rates.

Network topology

The basic issue in communication networks is the transmission of message to achieve a prescribed message throughput quality of service (QoS) and quantity of service. Depending on QoS, installation in the environment, economic considerations and the application, one or a combination of the

following topologies may be adopted: fully connected, mesh, star, ring, tree or bus. Of these, mesh type with some modifications has been found to be more useful (see Fig. 3(a)).

Mesh networks are regularly distributed networks that generally allow transmission only to a node's nearest neighbours. The nodes in these networks are generally identical, so the mesh networks are also referred to as peer-to-peer (see Fig. 2(a)). Mesh networks are good for large WSNs that are distributed over a geographic region, such as personnel or vehicle security surveillance systems. Another advantage of mesh networks is that, although all nodes are identical and have the same computing and transmission capabilities, certain nodes can be designed as 'group leaders' that take on additional functions (see Fig. 2(b)).

Multicast systems in mesh networks use a hierarchical leader based scheme for message transmission. Each group of nodes has a designated leader that is responsible for receiving messages from and transmitting to nodes outside the group. The group leader for communication should be the entry node of each group, while the group leader for DSP should be the exit node. Multicast routing improves efficiency and reduces message path length (see Fig. 2).

Routing tables for distributed networks increase exponentially as nodes are added, and hence the number of links increases exponentially. To simplify the network structure, one can use hierarchical clustering techniques (see Fig. 3(c)). The hierarchical structure must have the same structure at each level.

Hierarchical network structure simplifies routing and is also amenable to distributed signal processing and decision making, since some processing can be done at each hierarchical layer. Failure recovery is also straightforward. If a link fails, one may simply switch in one of the disabled links to take over. Hierarchical structure must be consistent, that is, it must have the same structure at each level.

Edge binding. The phenomenon of edge binding means that much of the routing power of peripheral stations is wasted because peripheral links are unused. Thus message tends to reflect the boundary into the interior or move parallel to periphery. To avoid this, the Manhattan geometry connects the nodes at one edge of the network to nodes at the opposite edge, thereby reducing the complexity (see Fig. 3(c)).

Power management. With the advent of ad hoc networks of geographically distributed sensors in remote site locations, there is a focus on increasing life time of sensor nodes through power generation and power conservation. Researchers are designing micro-electro-mechanical systems (MEMS), which can generate power using solar, vibration and thermal energies.

Miniature RF components for transceivers. RF identification (RFID) devices can also be used for power management purposes. The transponder micro-circuit has an L-C tank circuit which stores power from received interrogation signals, and uses that power to transmit a response. RFID operates in a low frequency range of 100kHz to 1.5MHz, or a high frequency range of 900MHz to 2.4GHz, and has an operating range up to 30 metres. TDMA is especially useful for power conservation, since a node can remain inoperative between its assigned time slots and wake up in time to receive and transmit messages.

The required transmission power increases as the square of the distance between source and destination. Therefore multiple short message transmission hops require less power than one long hop. A current topic of